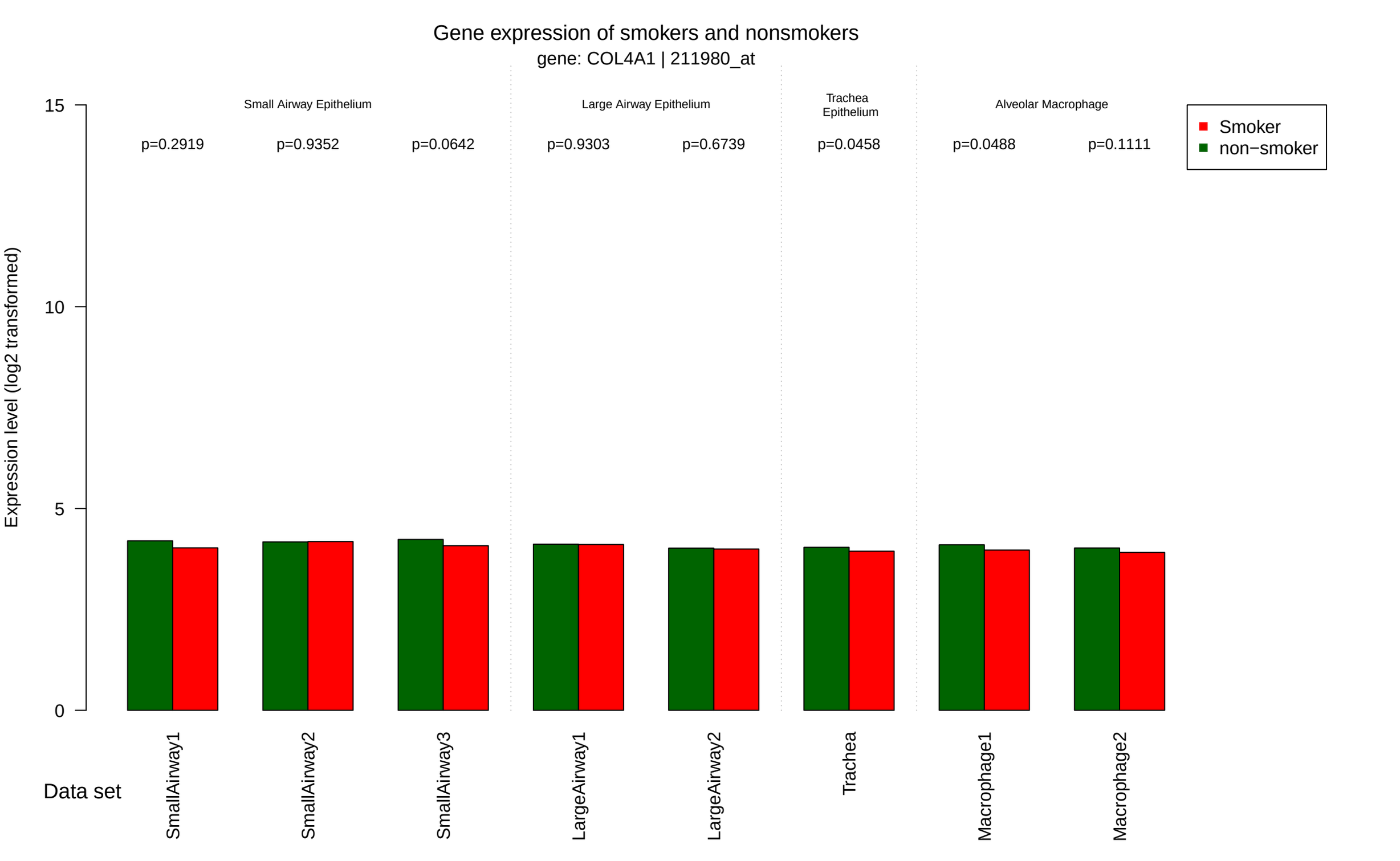
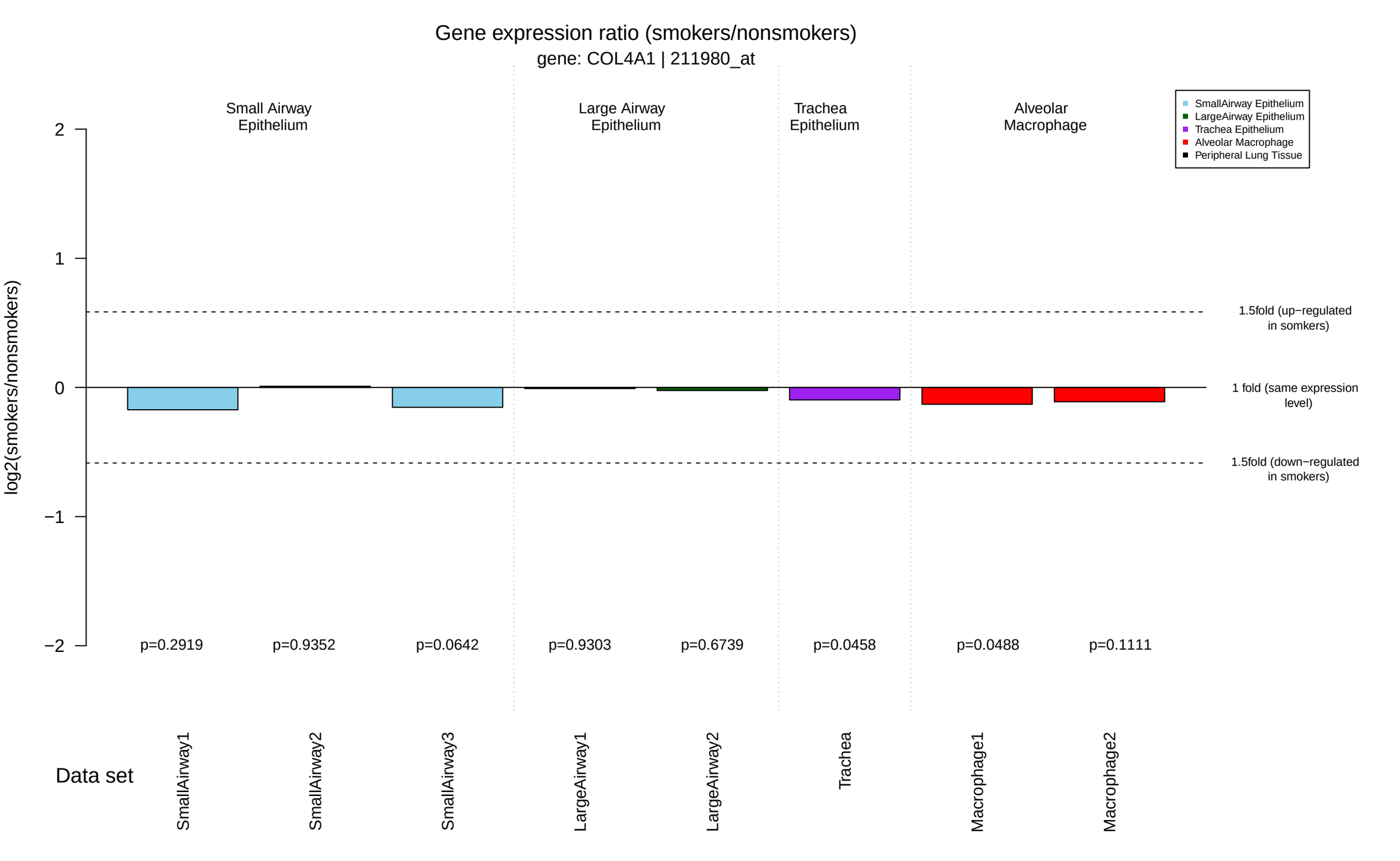
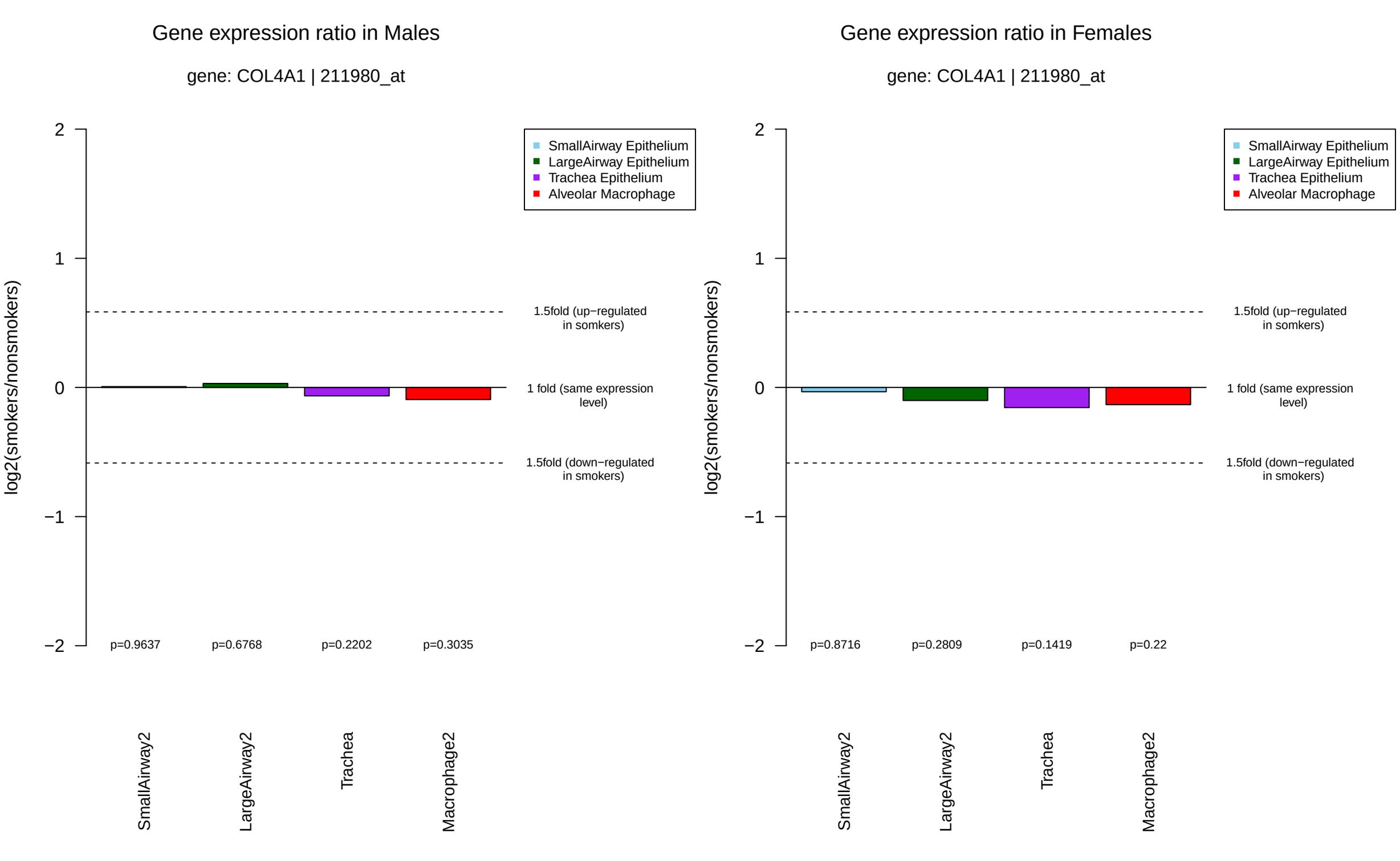
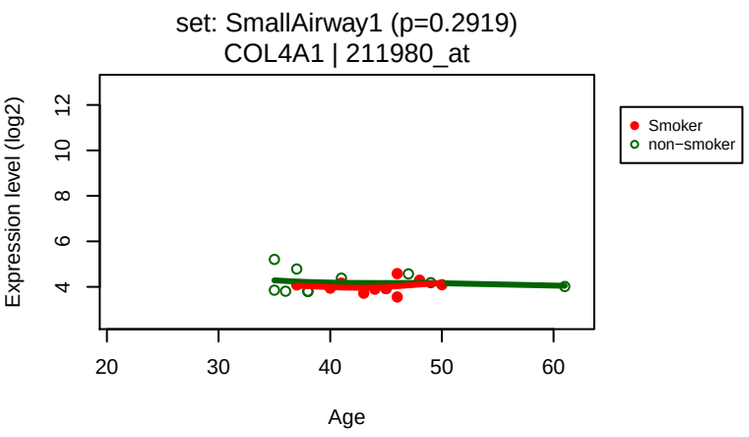
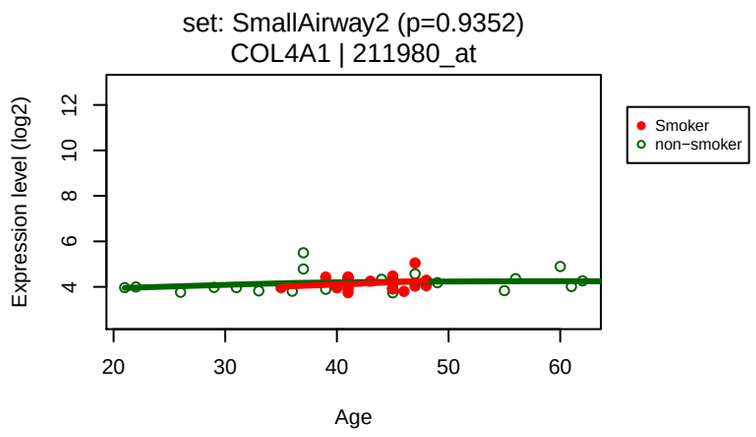
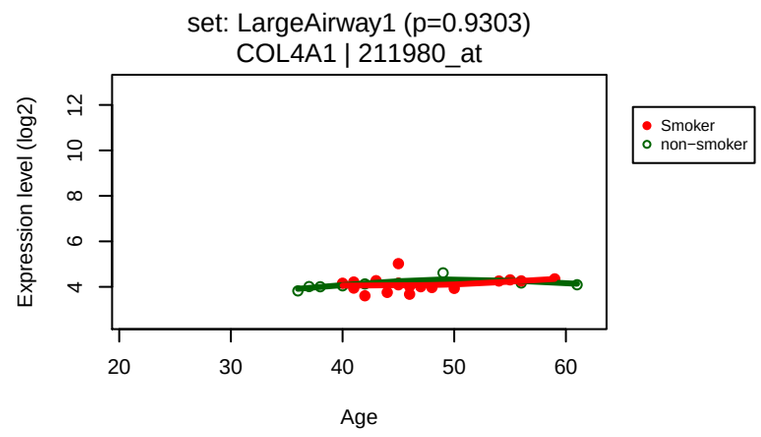

Gene expression of smokers and nonsmokers
gene: COL4A1 | 211980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2919
p=0.9352
p=0.0642
p=0.9303
p=0.6739
p=0.0458
p=0.0488
p=0.1111
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

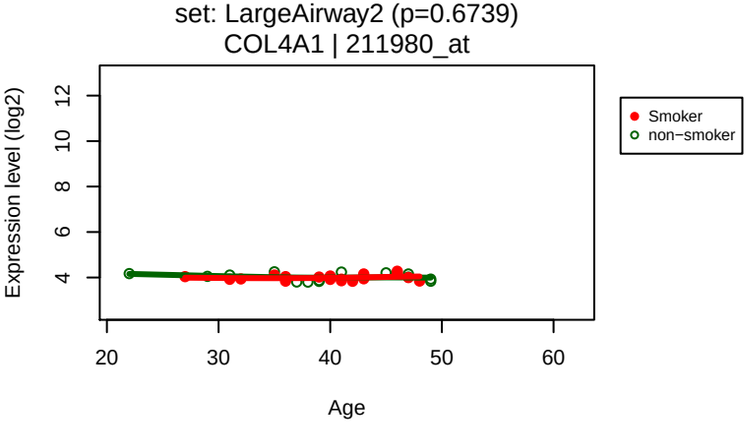

Gene expression ratio (smokers/nonsmokers)
gene: COL4A1 | 211980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2919
p=0.9352
p=0.0642
p=0.9303
p=0.6739
p=0.0458
p=0.0488
p=0.1111
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

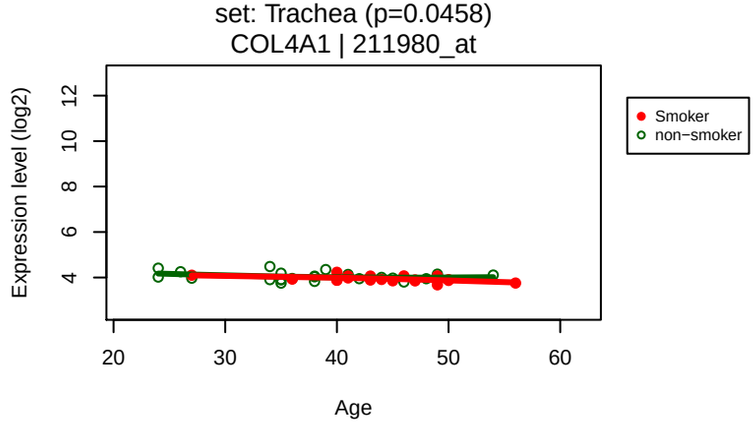

Gene expression ratio in Males
gene: COL4A1 | 211980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9637
p=0.6768
p=0.2202
p=0.3035
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A1 | 211980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8716
p=0.2809
p=0.1419
p=0.22
SmallAirway2
LargeAirway2
Trachea
Macrophage2

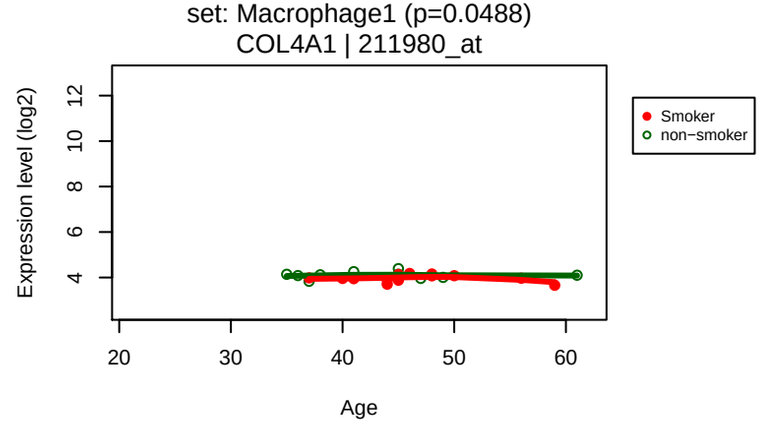

set: SmallAirway1 (p=0.2919)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker

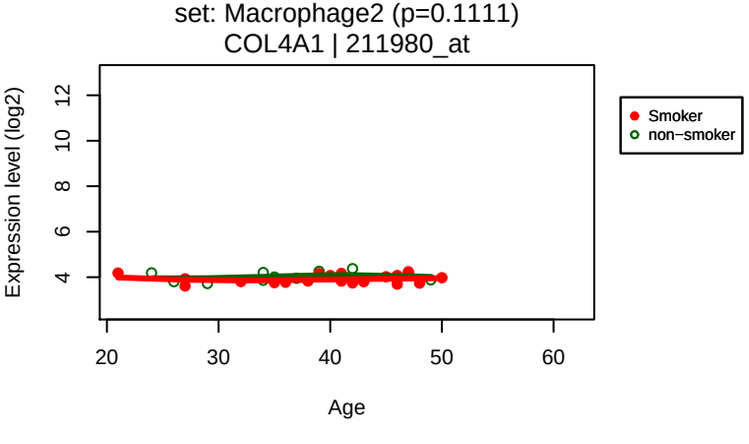

set: SmallAirway2 (p=0.9352)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker

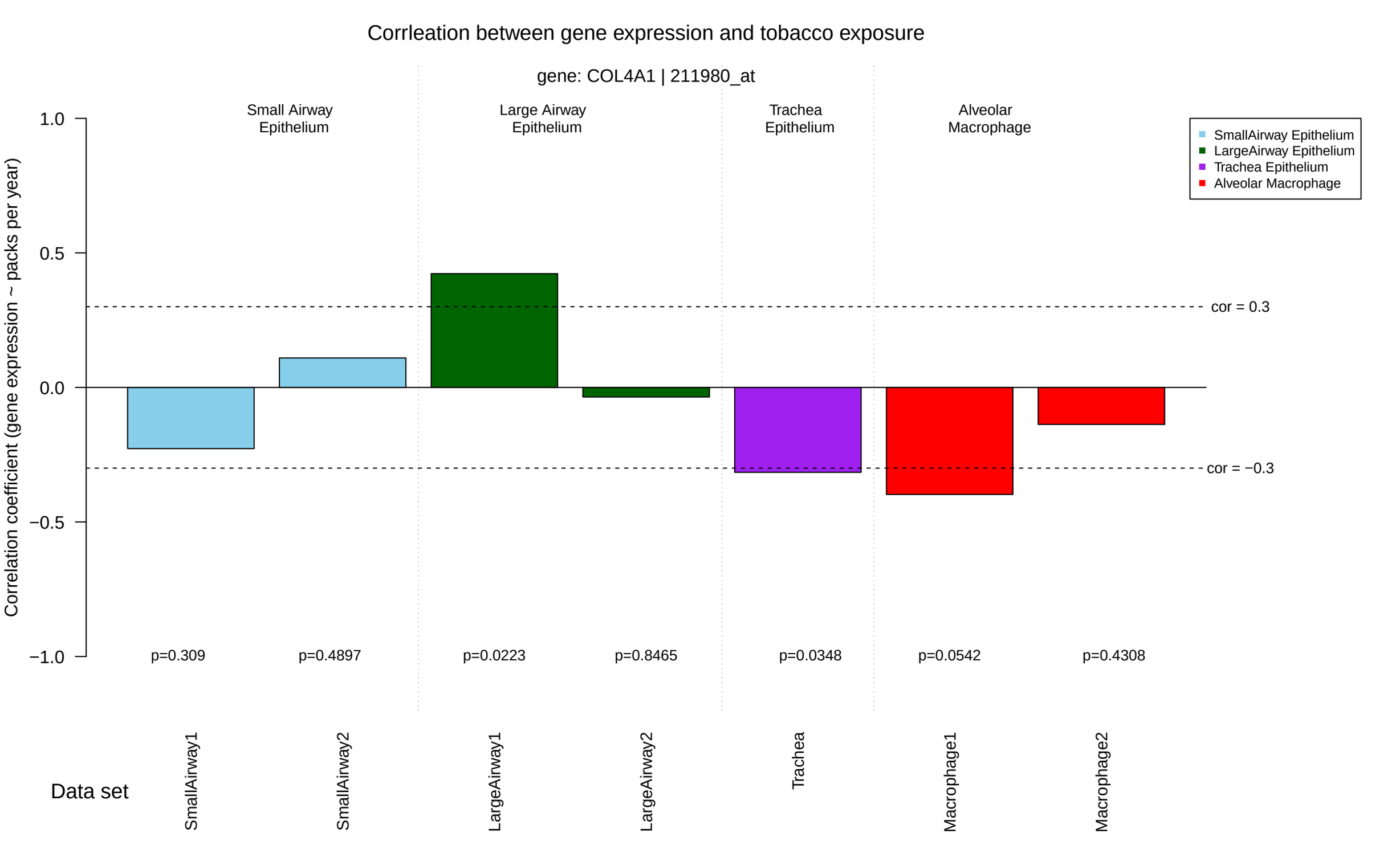

set: LargeAirway1 (p=0.9303)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6739)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0458)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0488)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1111)
COL4A1 | 211980_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A1 | 211980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.309
p=0.4897
p=0.0223
p=0.8465
p=0.0348
p=0.0542
p=0.4308
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2